Energy Preamble

Synopsis: What is energy? In this preamble, we explore the concept of energy through a set of activities. So far, we have studied motion (position, velocity, and acceleration) through the lens of forces – push and pull, tension, gravity, normal, and friction, to name a few. The key idea was that a net force causes a constant acceleration. Then, with a fixed known initial position and velocity, we can describe the motion completely at all times using kinematic equations.

There is another way and a different perspective to looking at motion: Energy or, more precisely, energy change. We can understand motion as objects' attempt to find the most stable energy state. Imagine a large rock precariously balancing on the tip of a tall mountain – you would call this an "unstable" situation or state. The rock will want to tumble down the hill, and as it does, it gains speed/momentum, eventually settling down at the foot of the mountain on a flat surface – now, you would call this a "stable" situation or state. You can describe the motion of the rock by saying the force of gravity accelerated the rock down the hill – a force perspective. Now, you can describe this same motion by saying the rock went from an unstable energy state to a stable energy state, so there was an energy change – an energy perspective. This "desire" to reach the minimum energy state is a powerful idea behind the motion of objects. You can think of every object in the universe like a rock on the "energy" landscape – it might be sitting at the top of a mountain about to tumble down (unstable), at the bottom of a hill (stable), or in between, on the side of a mountain heading down (in motion heading towards stability). This energy perspective has a beginning and end state, and the energy change between these two states. This energy change is the key to describing the motion, and you can guarantee that forces are present where there is a change.

When left to its own devices, an object will ALWAYS want to find the minimum or most stable energy state and, in doing so, "loses" or lowers its energy. However, an object can "gain" energy or raise its energy. For example, you can push a rock at the bottom of a valley back up to the top of the mountain, which will take a lot of energy. An input of energy now causes the upward-going motion. Recognize that to cause a motion that is uphill in the energy landscape (not "natural") costs energy; something needs to provide energy for that process.

In the following activities, I want you to think about the types of energy involved, the change between these types, and what role forces play.

Performance Expectations: Define energy, find different types of energy, and use the concept of energy to describe motion in activities. Describe the role of forces in energy change.

Copyright © 2026 by Dr. Byungmoon Cho

For PHY 151 — with calculus: Read the energy landscape as a potential energy function and recover the direction of the force from its slope.

Instructions and Questions

1) In your own words, define energy.

Your definition of energy	**Physics definition of energy**

2) Find out kinetic (KE), potential (PE), and thermal ("internal", TE) energies.

	What you found	**Physics definition**
KE		
PE		
TE		

3) The term "work" has a precise meaning in physics. What is it?

What you found	**Physics definition**

The landscape is a function.

Copyright © 2026 by Dr. Byungmoon Cho

The picture of a rock on a hillside is not just a metaphor. Draw the potential energy of the rock against its position and you get an actual curve, U(x) — the energy landscape — and "wants to roll downhill" has a precise meaning: the force on the rock is minus the slope of that curve, F = −dU/dx. Steep hillside, big force. Flat valley floor, no force at all, which is why it stops there. We will make this exact in Note 5; for now, keep the picture and the sign.

i) Sketch a landscape U(x) with one hill and one valley. Mark where F = 0, and say which way F points on each side of the hill.

ii) Where on your sketch is the rock "unstable", in the language of the synopsis? What is true of the slope there?

4) Perform activities. As you do, identify the object of interest, sketch a diagram showing forces and velocities (if applicable), and indicate various energies involved.

i) Jump up on the desk. Jump back down.

Object of interest: When/where do you have KE: PE: TE:	sketch diagram

ii) Toss a ball up and catch it.

Object of interest: When/where do you have KE: PE: TE:	sketch diagram

iii) Toss a ball across the room.

Copyright © 2026 by Dr. Byungmoon Cho

Object of interest: When/where do you have KE: PE: TE:	sketch diagram

iv) Push a classmate on a trolley.

Object of interest: When/where do you have KE: PE: TE:	sketch diagram

v) Roll a steel marble down an incline.

Object of interest: When/where do you have KE: PE: TE:	sketch diagram

vi) Propel a steel marble up an incline.

Copyright © 2026 by Dr. Byungmoon Cho

Object of interest: When/where do you have KE: PE: TE:	sketch diagram

Copyright © 2026 by Dr. Byungmoon Cho